Companions hidden in public HARPS velocities: new stellar companions and a dark 1.2 $M_\odot$ companion to the giant HD 20037

Charles C. Norton

Abstract

HARPS-RVBank holds 212 562 public velocities of 2 912 stars, and Gaia DR3 measures the same stars' motions on the sky. Searched together, they reveal companions that no catalogue holds. A census of the velocities for variation beyond the errors, set against the NASA Exoplanet Archive, SB9 and Gaia DR3's two-body orbits, followed by Keplerian fits sampled with emcee and inclinations bounded by Gaia's astrometry, finds stellar and substellar companions of HD 107181, HD 77462, HD 7399 and HD 134985 with no published orbit. The G8 giant HD 20037, at 2.90 $M_\odot$, carries a companion whose orbit follows from its HARPS velocities, Gaia DR3's velocity-trend and acceleration solutions and its Hipparcos-Gaia proper-motion anomaly fitted together: a period of $28.5^{+3.2}_{-2.5}$ yr, an eccentricity of 0.46 ± 0.04, an inclination of $98.8 \pm 1.2°$ and a mass of $1.18 \pm 0.05\,M_\odot$. Its lines do not appear in the giant's spectra along its own track, where a main-sequence star of that mass, at 4 per cent of the light, would read at 7σ, so the companion is a massive white dwarf or a close pair of M dwarfs.

Keywords: Techniques: radial velocities, Astrometry, Binaries: spectroscopic, White dwarfs

1. Introduction

Radial-velocity surveys have watched thousands of nearby stars for two decades, and their archives hold more than their programmes published. HARPS-RVBank [1] recomputed every public HARPS velocity with SERVAL [2], corrected the instrument's nightly zero points, and attached each spectrum's activity indicators, so that a star's whole record can be read at once. Gaia DR3 adds two independent views of the same stars: its own spectroscopic velocities and their variability, with orbital and trend solutions for some, and astrometry whose acceleration and proper-motion anomaly against Hipparcos [3] bound a companion's mass and inclination [4].

This paper reads the two together for companions that no catalogue holds. Section 2 describes the data, Section 3 the stellar and substellar companions and Section 4 the dark companion of HD 20037.

2. Data

The velocities are HARPS-RVBank's SERVAL velocities with their nightly zero points corrected, 2 912 stars and 212 562 velocities, with the chromatic index, differential line width, Hα and Na D indices and the pipeline's line width and bisector of each spectrum. Each star is matched to Gaia DR3 [4] as the brightest source within 30 arcsec of its J2000 position whose G lies within 2.5 mag of its V, which follows stars of high proper motion to Gaia's epoch, and to the Hipparcos-Gaia proper-motion anomaly of ref. [3] through its Hipparcos number. The public HARPS spectra

Table 1: Companions with no published orbit: the velocities (nights), the orbit's period, semi-amplitude and eccentricity with their 16th to 84th percentiles, the companion's minimum mass for the primary's mass m_1, and Gaia's RUWE and the Hipparcos-Gaia proper-motion anomaly's signal to noise.

Star	Type	Velocities (nights)	P (d)	K ($\mathrm{km\,s^{-1}}$)	e	m_1 ($\mathrm{M_\odot}$)	$m_2 \sin i$ ($\mathrm{M_\odot}$)	RUWE, PMa S/N
HD 107181	G3 V	16 (11)	13.0	15.6 (15.0–16.2)	0.10 (0.06–0.14)	1.03	0.196 (0.187–0.204)	1.13, 38.9
HD 77462	F7 V	11 (11)	345 (338–355)	4.57 (4.37–4.85)	0.47 (0.45–0.49)	1.30	0.172 (0.165–0.183)	14.0, 5.8
HD 7399	F5 V	38 (22)	524 (493–558)	0.52 (0.51–0.55)	0.07 (0.04–0.11)	1.23	0.023 (0.022–0.025)	1.87, 15.1
HD 134985	K0 V	23 (23)	8700 (7000–11000)	0.59 (0.41–0.76)	0.37 (0.32–0.42)	0.90	0.050 (0.034–0.073)	1.17, 96.3

of the stars discussed here come from the ESO Science Archive, and their ultraviolet and X-ray fluxes from GALEX [5], eROSITA's first survey [6], ROSAT's 2RXS [7] and 4XMM.

3. Stellar and substellar companions

A star enters the census when its velocities vary beyond their errors (a reduced χ^2 above 25, a scatter above $30\,\mathrm{m\,s^{-1}}$ and above five times the errors by the median absolute deviation, on four or more nights), when the SERVAL velocities agree with the pipeline's (correlation above 0.9), and when it is of spectral type F, G, K or M and luminosity class III or fainter. Of the 2 912 stars, 240 vary; 100 of them are explained by the NASA Exoplanet Archive, SB9 or a Gaia DR3 two-body orbit and 76 more are known binaries, variables or substellar hosts in SIMBAD. The 32 of the remaining 64 with ten or more nights were fitted with a Keplerian, and the thirteen strongest measured with a Keplerian, a linear trend and a jitter added to the errors, sampled with emcee [8], the periodogram's false-alarm level from shuffled velocities. Gaia bounds each orbit's inclination: for each of 200 orbits from the chain, 60 inclinations uniform in $\cos i$ and 24 nodes, the primary's photocentre, the companion taken dark, is followed through the Hipparcos and Gaia DR3 windows, its predicted proper-motion anomaly set against the observed one for periods beyond 300 d, and its predicted wobble over Gaia's window, at 0.3 mas a transit, against the RUWE. SIMBAD's references and the papers they cite give each star's literature.

Four companions have no published orbit (Table 1). HD 107181, a G3 dwarf, has a companion of at least $0.196\,\mathrm{M_\odot}$ on a 13.0 d orbit, and its velocities scatter by $1.2\,\mathrm{km\,s^{-1}}$ about that orbit while its proper-motion anomaly stands at 39σ, which the 13-day orbit cannot produce: a third body lies farther out. HD 77462, an F7 dwarf, has a companion of at least $0.172\,\mathrm{M_\odot}$ on an eccentric 345 d orbit, and its RUWE of 14.0 shows the star's wobble on the sky. HD 7399, an F5 dwarf, moves on a 524 d orbit at only $0.52\,\mathrm{km\,s^{-1}}$, a minimum mass of $24\,\mathrm{M_J}$ in the brown-dwarf desert; Gaia DR3 holds no orbit for it but an acceleration solution (Acceleration9, significance 26), and with its RUWE of 1.87 and proper-motion anomaly of 15σ the star moves more on the sky than $24\,\mathrm{M_J}$ at 524 d would move it, either because this companion is seen nearly face on, near $0.18\,\mathrm{M_\odot}$, or because a further body pulls the star. HD 134985 (GJ 579.3), a K0 dwarf, has a companion of at least $52\,\mathrm{M_J}$ on an orbit of about 24 yr, partly covered by its 8.3-year series, with a proper-motion anomaly of 96σ.

4. The dark companion of HD 20037

HD 20037 is a G8 giant of $2.92 \pm 0.15\,\mathrm{M_\odot}$ and $10.3\,\mathrm{R_\odot}$ [3] at 172 pc. Its 13 HARPS velocities fall smoothly by $3.3\,\mathrm{km\,s^{-1}}$ from 2006 to 2011 with nights a day apart agreeing to a few metres per second. Gaia DR3 holds three more views of the same orbit. Its 27 velocities over 2014 to 2017

Table 2: HD 20037's orbit from its HARPS velocities and Gaia's and Hipparcos's astrometry together: medians with 16th to 84th percentiles of 4 000 samples.

Period (d)	10 423 (9 511–11 609)
Eccentricity	0.46 (0.43–0.51)
Giant's semi-amplitude K_1 ($\mathrm{km\,s^{-1}}$)	5.00 (4.95–5.11)
Inclination (deg)	98.8 (97.6–100.0)
Semi-major axis (AU)	14.9 (14.0–16.1)
Giant's mass ($\mathrm{M_\odot}$)	2.90 (2.76–3.07)
Companion's mass ($\mathrm{M_\odot}$)	1.18 (1.14–1.23)

vary beyond their errors and are fitted by a second-degree trend (SecondDegreeTrendSB1) rising at $0.0071 \pm 0.0036\,\mathrm{km\,s^{-1}}$ per day at 2016.0; its astrometry carries an acceleration solution (Acceleration7) at a significance of 31.9, -1.468 ± 0.046 and $-0.705 \pm 0.047\,\mathrm{mas\,yr^{-2}}$ in right ascension and declination, with the proper motion at 2016.0; and its mean motion between Hipparcos and Gaia differs from that proper motion at 114σ [3].

One Keplerian orbit of the giant about the barycentre, the companion dark, is fitted to all of it at once. The HARPS velocities are read directly with a zero point and a jitter. Gaia's velocity derivatives are those of a quadratic fitted to the model's velocities at 27 times across its 968-day velocity window; the proper motion and acceleration at 2016.0 are those of a quadratic fitted to the photocentre over Gaia's astrometric window; and the long-term mean motion is the photocentre's displacement from its Hipparcos position, a line fitted over Hipparcos's window, to its Gaia position over the 24.75 yr between them. The parameters are the period, the time of periastron, $\sqrt{e}\cos\omega$, $\sqrt{e}\sin\omega$, the companion's mass, $\cos i$, the node, the HARPS zero point, the jitter and the giant's mass with its prior. The posterior was located by differential evolution from four starts and sampled with emcee. Every data set is met (Table 2): the proper-motion anomaly to within $0.03\,\mathrm{mas\,yr^{-1}}$, the acceleration to within 1σ, Gaia's velocity slope at $0.0080\,\mathrm{km\,s^{-1}}$ per day, and the HARPS velocities to $13\,\mathrm{m\,s^{-1}}$. The companion weighs $1.18 \pm 0.05\,\mathrm{M_\odot}$ on an orbit of 28.5 yr seen nearly edge on; periastron passed in 2013, between the HARPS and Gaia series.

A main-sequence star of $1.18\,\mathrm{M_\odot}$, an F8 or G0 dwarf, would give about 4 per cent of the system's V-band light. With the orbit known, the companion's lines can be sought along their own track: in each spectrum it moves against the giant at $-(1 + m_1/m_2)$ times the giant's velocity about the systemic one, here +5.0 to $+16.6\,\mathrm{km\,s^{-1}}$. The spectra are placed in the giant's frame on a grid uniform in velocity, divided by their median and whitened, cross-correlated with the line depths of a G0 dwarf, the rows' common pattern taken out, and read along that track. The track reads $z = -0.3$ against 2 000 shuffles of the rows' velocities ($p = 0.61$), while the same G0 spectrum laid into the actual spectra along the track at 0.5, 1, 2 and 4 per cent of the light reads $z = 1.6$, 3.7, 6.2 and 7.3. A single main-sequence companion of this mass is therefore excluded. GALEX measures HD 20037 at FUV 19.76 ± 0.11 and NUV 14.24 ± 0.01, a G8 giant's colours with no hot component. eROSITA records a soft source 1.8 arcsec from the star, 17 counts mostly between 0.5 and 1 keV, $2.0 \times 10^{29}\,\mathrm{erg\,s^{-1}}$ in the 0.2 to 2.3 keV band, as a giant's corona or an active pair of M dwarfs would give and far below a neutron star accreting the giant's wind.

The companion is therefore dark at $1.18\,\mathrm{M_\odot}$: a white dwarf in the upper few per cent of the mass distribution, whose progenitor was more massive than the giant is now, or a close binary of two M dwarfs near $0.6\,\mathrm{M_\odot}$ each, whose combined light, below 0.2 per cent of the giant's, the spectra cannot reach. The eROSITA source fits the second, but a G8 giant can supply it alone.

5. Conclusion

Public HARPS velocities read beside Gaia DR3's astrometry and velocities give four stellar and substellar companions with no published orbit, among them a 13-day M dwarf in a triple and a brown-dwarf-desert minimum mass that the astrometry turns into a low-mass star. They give the giant HD 20037 a dark companion of $1.18 \pm 0.05\,\mathrm{M}_\odot$ on a 28.5-year orbit, whose lines are absent where a main-sequence star of its mass would stand at 7σ: a massive white dwarf or a pair of M dwarfs.

Acknowledgements

Based on data obtained from the ESO Science Archive Facility, and on HARPS-RVBank. This work has made use of data from the European Space Agency mission Gaia, processed by the Gaia Data Processing and Analysis Consortium, funded by national institutions, in particular the institutions participating in the Gaia Multilateral Agreement; of GALEX and eROSITA data; of the NASA Exoplanet Archive, operated by the California Institute of Technology under contract with NASA; and of the SIMBAD database and the VizieR catalogue access tool, operated at CDS, Strasbourg, France.

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